Universidad Tecnológica del Perú

Cálculo II

taller 1

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Ejercicio 1

Determine si las siguientes integrales convergen o divergen

1.1

$$\int_3^7 \frac{2}{x-3}dx$$

$$x \in]3,7]$$

$$\lim_{a\to 3^+} \int_a^7 \frac{2}{x-3}dx$$

$$\lim_{a\to 3^+} 2\ln|x-3|\Big|_a^7$$

$$\lim_{a\to 3^+} 2(\ln|4| - \ln|a-3|)$$

$$\lim_{a\to 3^+} 2(\ln|4| - (-\infty))$$

$$+\infty$$

La integral diverge, ya que el límite es infinito.

1.2

$$\int_0^{+\infty} 5e^{-2x}dx$$

$$\lim_{a\to \infty^+} \int_0^a 5e^{-2x}dx$$

$$\left[\frac{5e^{-2x}}{-2}\right]_0^a$$

$$\left[\frac{5e^{-2a}}{-2} - \left(\frac{5e^0}{-2}\right)\right]$$

$$\left[-\frac{5e^{-2a}}{2} + \frac{5}{2}\right]$$

$$\lim_{a\to \infty^+} \left(0 + \frac{5}{2}\right) \frac{5}{2}$$

La integral converge, ya que el límite es un valor finito.

1.3

$$\int_0^{+\infty} 3xe^{-x^2}dx$$

$$\int_0^{+\infty} 3xe^{-x^2}dx$$

$$u = -x^2$$

$$du = -2xdx$$

$$\int_0^{+\infty} 3xe^{-x^2}dx$$

$$\int_0^{+\infty} -\frac{3}{2}e^u du$$

$$\left[-\frac{3}{2}e^u\right]_0^{+\infty}$$

$$\left[-\frac{3}{2}e^{-\infty} + \frac{3}{2}e^0\right]$$

$$\left[0 + \frac{3}{2}\right]$$

$$\frac{3}{2}$$

Ejercicio 2

Usando la definición de función Gamma evalúe las siguientes integrales

2.1

$$\int_0^{+\infty} e^{-10x}\sqrt{x^5}dx$$

$$t = 10x$$

$$x = \frac{t}{10}$$

$$dx = \frac{dt}{10}$$

$$\int_0^{+\infty} x^{\frac{5}{2}}e^{-10x}dx$$

$$\int_0^{+\infty} \left(\frac{t}{10}\right)^{\frac{5}{2}} e^{-t}\frac{dt}{10}$$

$$\int_0^{+\infty} \frac{t^{\frac{5}{2}}}{10^{\frac{5}{2}}}e^{-t}\frac{dt}{10}$$

$$\int_0^{+\infty} \frac{t^{\frac{5}{2}}}{10^{\frac{7}{2}}}e^{-t}dt$$

$$\frac{1}{10^{\frac{7}{2}}}\int_0^{+\infty} t^{\frac{5}{2}}e^{-t}dt$$

$$\frac{1}{10^{\frac{7}{2}}}\cdot\Gamma\left(\frac{5}{2}+1\right)$$

$$\frac{1}{10^{\frac{7}{2}}}\cdot\Gamma\left(\frac{7}{2}\right)$$

$$\frac{1}{10^{\frac{7}{2}}}\cdot\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\sqrt{\pi}$$

$$\frac{1}{10^{\frac{7}{2}}}\cdot\frac{15}{8}\sqrt{\pi}$$

2.2

$$\int_0^{+\infty} \frac{\sqrt{x^9}}{e^{3x}} dx$$

$$t = 3x$$
$$x = \frac{t}{3}$$
$$dx = \frac{dt}{3}$$
$$\int_0^{+\infty} \left(\frac{t}{3}\right)^{\frac{9}{2}} e^{-t}\frac{dt}{3}$$
$$\int_0^{+\infty} \frac{t^{\frac{9}{2}}}{3^{\frac{9}{2}}} e^{-t}\frac{dt}{3}$$
$$\int_0^{+\infty} \frac{t^{\frac{9}{2}}}{3^{\frac{11}{2}}} e^{-t}dt$$
$$\frac{1}{3^{\frac{11}{2}}}\int_0^{+\infty} t^{\frac{9}{2}} e^{-t}dt$$
$$\frac{1}{3^{\frac{11}{2}}}\cdot \Gamma\left(\frac{9}{2}+1\right)$$
$$\frac{1}{3^{\frac{11}{2}}}\cdot \Gamma\left(\frac{11}{2}\right)$$
$$\frac{1}{3^{\frac{11}{2}}}\cdot \frac{9}{2}\cdot\frac{7}{2}\cdot\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\sqrt{\pi}$$
$$\frac{1}{3^{\frac{11}{2}}}\cdot \frac{945}{32}\sqrt{\pi}$$

Ejercicio 3

Usando la definición de funciń Beta evalúe las siguientes integrales

3.1

$$\int_0^5 \frac{x^5}{\sqrt{5-x}} dx$$

$$x \in [0,5[$$
$$\frac{x}{5} = t$$
$$x = 5t$$
$$dx = 5dt$$
$$\int_0^5 (5t)^5 \cdot (5-x)^{-\frac{1}{2}} \cdot 5dt$$
$$\int_0^1 5^5 \cdot t^5 \cdot (5-5t)^{-\frac{1}{2}} \cdot 5dt$$
$$5^5 \cdot 5^1 \int_0^1 t^5(5-5t)^{-\frac{1}{2}} \cdot dt$$
$$5^6 \int_0^1 t^5 \cdot 5^{-\frac{1}{2}} \cdot (1-t)^{-\frac{1}{2}} \cdot dt$$
$$5^{\frac{11}{2}} \int_0^1 t^5(1-t)^{-\frac{1}{2}}dt$$
$$5^{\frac{11}{2}} \cdot B(5+1, -\frac{1}{2}+1)$$
$$5^{\frac{11}{2}} \cdot B(6, \frac{1}{2})$$
$$5^{\frac{11}{2}} \cdot \frac{\Gamma(6)\cdot\Gamma(\frac{1}{2})}{\Gamma(\frac{13}{2})}$$
$$\frac{5^{\frac{11}{2}}\cdot 5! \cdot \sqrt{\pi}}{\frac{11}{2}\cdot\frac{9}{2}\cdot\frac{7}{2}\cdot\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\sqrt{\pi}}$$

3.2

$$\int_0^8 \frac{x^3}{\sqrt{8-x}} dx$$

$$x \in [0, 8[$$

$$\frac{x}{8} = t$$

$$x = 8t$$

$$dx = 8dt$$

$$\int_0^8 (8t)^3 \cdot (8-x)^{-\frac{1}{2}} \cdot 8dt$$

$$\int_0^1 8^3 \cdot t^3 \cdot (8-8t)^{-\frac{1}{2}} \cdot 8dt$$

$$8^3 \cdot 8^1 \int_0^1 t^3 (8-8t)^{-\frac{1}{2}} \cdot dt$$

$$8^4 \int_0^1 t^3 \cdot 8^{-\frac{1}{2}} \cdot (1-t)^{-\frac{1}{2}} \cdot dt$$

$$8^{\frac{7}{2}} \int_0^1 t^3 (1-t)^{-\frac{1}{2}} dt$$

$$8^{\frac{7}{2}} \cdot B(3+1, -\frac{1}{2}+1)$$

$$8^{\frac{7}{2}} \cdot B(4, \frac{1}{2})$$

$$8^{\frac{7}{2}} \cdot \frac{\Gamma(4) \cdot \Gamma(\frac{1}{2})}{\Gamma(\frac{9}{2})}$$

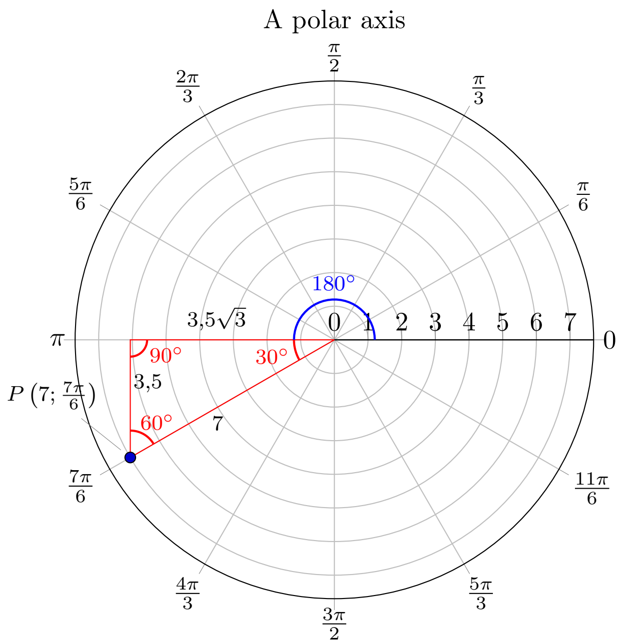
$$\frac{8^{\frac{7}{2}} \cdot 3! \cdot \sqrt{\pi}}{\frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}}$$

Ejercicio 4

Dada las siguientes coordenadas cartesianas, determine sus coordenadas polares y grafíque-las en el plano polar

4.1

$$(-3{,}5\sqrt{3}; -3{,}5)$$



$$210 \cdot \frac{\pi}{180} = \frac{7\pi}{6}$$

4.2

$$(12; -12)$$